Q. Constructor in Java

A **constructor** in Java is a **special method used to initialize an object** when it is created.

When you create an object of a class, the constructor **automatically runs** and assigns initial values to variables.

Key Points

1. The constructor name **must be the same as the class name**.
2. It **does not have a return type** (not even void).
3. It is **automatically called when an object is created**.
4. It is mainly used to **initialize instance variables**.

1. Simple Example (Default Constructor)

```
class Student
{
   int id;
   String name;

   Student()   // Constructor
   {
      id = 101;
      name = "Rahul";
   }

   void display()
   {
      System.out.println(id + " " + name);
   }

   public static void main(String args[])
   {
      Student s1 = new Student();   // constructor called
      s1.display();
   }
}
```

Output : 101 Rahul

2. Constructor with Parameters (Parameterized Constructor)

Sometimes we want **different values for different objects**.

```
class Student
{
   int id;
   String name;

   Student(int i, String n)   // Parameterized constructor
   {
      id = i;
      name = n;
   }

   void display()
   {
```

```
        System.out.println(id + " " + name);
    }

    public static void main(String args[])
    {
        Student s1 = new Student(101, "Rahul");
        Student s2 = new Student(102, "Amit");

        s1.display();
        s2.display();
    }
}
```

Output: 101 Rahul
102 Amit

3. Why Constructors Are Important

Without constructors, you would need to assign values like this:

```
Student s1 = new Student();
s1.id = 101;
s1.name = "Rahul";
```

Constructors make object creation **clean and automatic**.

4. Types of Constructors in Java

Type	Meaning
Default Constructor	Constructor without parameters
Parameterized Constructor	Constructor with parameters

Q. Differentiate between Constructor and Method in Java.

<u>**Constructor**</u>	<u>**Method**</u>
1. A constructor is a special member of a class used to initialize objects.	**1.** A method is a member function used to perform a specific task.
2. The name of the constructor must be the **same as the class name**.	**2.** A method can have **any valid name**.
3. A constructor **does not have any return type**, not even `void`.	**3.** A method **must have a return type** such as `int`, `void`, `String`, etc.
4. Used to **initialize the object** at the time of object creation.	**4.** Used to **perform operations or define behavior** of an object.
5. It is **automatically called** when an object is created.	**5.** It is **called explicitly** using an object.
6. Executes **once when the object is created**.	**6.** Can be **called many times** during program execution.
7. Constructors **are not inherited**.	**7.** Methods **can be inherited** by subclasses.